

Himadri Mandal

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Education

École Normale Supérieure, Paris September 2026
Incoming Master's Student in Mathematics

Indian Statistical Institute, Kolkata August 2023 — May 2026
Statistics Undergraduate

Programming experience

Python: PyTorch, Numpy, Pandas, Bash, R, Next.js + TailwindCSS, $\text{\LaTeX}$, Linux (Arch Linux on i3)

Work experience

Researcher: Truth Audit Labs, Delhi June 2026 — Ongoing

Visiting Student Researcher: SCDLDS, Ashoka University May 2025 — January 2026

Research

Generalizing Hopf-Lax in Minmax Games: Sushant Vijayan Ongoing (**TAL**)
Using nonexpansive operators in hemi-metric spaces to generalize Hopf-Lax for minmax games with a subadditive initial function.

Fourier Entropy Influence Conjecture: Subhash Khot, **Courant** [Blog](#) — Ongoing

Hypercontractivity of Mixing Measures with Uncountable Spin States: Ayan Ghosh Paper

Does Order Matter: Robust Generalization: Mihir, Aritra et. al. [Submitted to NeurIPS 2026](#)

Active Simple Hypothesis Testing: Sushant Vijayan, **TIFR** [IEEE ISIT 2026](#)

Limits of Wtd Emp Samplers: Sarvesh R. Iyer, Sandeep Juneja, et. al. [Poster, AISTATS 2026](#)

Guarantees for Diffusion on Tilted Samples: Sandeep Juneja, et. al. [arXiv](#)

Polynomial Sparsity Bounds: A. R. Chowdhury, V. Bhargava (**IIT Bombay**) May 2025
Research project on improving the $d^2 \log(n)$ bound in the Sparsity Bounds conjecture.

Circuit Phenomenology Using Sparse Autoencoders: David Udell, **SPAR** [Preprint](#)

Selected Courses/Workshops

IEEE International Symposium of Information Theory, 2026: Guangzhou, China July 2026

exSPLORe2026 Workshop: SCDLDS, Ashoka University January 2026

DS: Probability and Optimization Methods: ICTS, Bangalore August 2025

exSPLORe2025 Workshop: SCDLDS, Ashoka University January 2025

CaMLAB: Cambridge AI Safety Hub April 2024

Projects

Theoretical.....

Independence Is Almost Dependence:

[Blog.](#)

Axiom of Choice, the Zorn's Lemma and their equivalence:

[Blog.](#)

Cold Reflections:

[Link.](#) My blog on Mathematics, Statistics, Philosophy and everything else that interests me. I add some of my work there.

Empirical/Programming.....

Ponderings on OthelloGPT:

[Blog.](#)

ORIGAMI:

[Repo.](#)

Selected fellowships and awards

ENS Paris Normalien: awarded April 2026, 1000€ per month (out of 4 selected internationals)

ISI K. Outstanding Performance: awarded January 2024, received ₹1500

ISI Kolkata B.Stat. Entrance: awarded August 2023, ranked 11th (out of 10,000 students)

Indian Olympiad Qualifier for Mathematics, KV: awarded February 2021, rank 4 in my region (out of >100,000 students all over India)

Atlas Fellowship India Finalist: awarded September 2022, received 1000\$, top 200 (out of 4000 students) in a rationality fellowship

IISc Enumeration Finalist: awarded October 2022

CMI Tessellate Finalist: awarded October 2021

More experience

Volunteer and outreach.....

Owner: awas

October 2020 — October 2022

Served as the **organizer and mentor** for daily math problem solving sessions, philosophical debates, and programming discussions for over two years on **Discord**. Mentored smart math enthusiasts, from all over India, learn hard math, and guided a few of them to get into the Indian training camp of IMO.

Owner: LML: Learning Machine Learning, ISI Kolkata

January 2024 — March 2025

Organized extensive discussions on and around Machine Learning (Theory & Applied), Artificial Intelligence, and Statistics papers for more than a year.

Core Member: Theory CS Club, ISI Kolkata

March 2025 — Ongoing

Engaged in extensive discussions on research papers on and around Theoretical CS focussing on Complexity Theory.